

Felix Borck

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PROFILE

I care about how knowledge works - in logic, language, representation, and now in language models. Since 2018, I have pursued that question through philosophy, provability theory, and vagueness. Since late 2025, I have developed it into AI-enabled ML research, using AI itself to generate and test empirical evidence about how meaning-like behavior is governed inside transformers and whether mechanistic explanations of that behavior survive intervention.

RESEARCH EXPERIENCE

Independent ML Researcher

Late 2025–present

Developed Appearance of Meaning (AoM), a framework for identifying which internal states causally govern controlled meaning-like behavior in transformer language models.

Developed Mechanics of Meaning (MoM), a framework for testing whether proposed feature-level explanations preserve that behavior under matched intervention.

Established a sufficiency result showing that purely distributional transformer states can causally govern a controlled meaning-like competence profile, and a limitation result showing that strong reconstruction fidelity alone does not validate sparse feature-level causal explanation.

Fundamental Researcher in Knowledge, Language, Mathematics, and Metaphysics

2018–2025

Conducted long-horizon, self-funded independent research in metaphysics, philosophy of language, logic, and formal structure, pursuing foundational questions beyond standard curricular frameworks.

Knowledge and provability: studied Godelian self-reference, Lob's theorem, GL, and Solovay's completeness result in relation to formal provability in Peano Arithmetic.

Meaning and its boundaries: in work on vagueness (MA thesis, grade 1.4), critically examined Williamson's epistemic theory by arguing that its instability-based solution blocks Sorites synchronically but generates a diachronic instability that threatens the classical stability it aims to preserve, and extended the analysis to concept boundaries and uncertainty in language models.

Structure of representation: investigated foundational conditions under which representational structures - formal, linguistic, conceptual - acquire explanatory power.

Founder and Research Engineer, Sapphire Bridge

2024–present

Built a regulation-constrained RAG system for citation-heavy financial reporting.

Raised angel investment and public grants for a 2025 AI product venture.

Built Voice Post Pilot, a voice-to-social-media SaaS product, on a solo-built AWS serverless stack (Lambda, CloudFront, Route 53), launched in 16 weeks.

PUBLICATIONS AND MANUSCRIPTS

The Appearance of Meaning: Context-Dependence and Semantic Competence in Transformer Architectures. Under review at the Journal of Logic, Language and Information (2026).

Mechanics of Meaning: Sparse Feature Interventions and the Basis Structure of Contextual Control in Transformers. Completed manuscript (2026).

Geometric Fidelity Is Not Behavioral Preservation. Completed manuscript, draft available on request (2026).

Context-Dependent Semantics in Large Language Models: A Case for Transformer Architectures. ESSLLI 2025 Student Session.

Notwendigkeit als Beweisbarkeit. Research paper, Goethe University Frankfurt (2019).

METHODS AND AREAS

ML. Mechanistic interpretability, causal intervention, LLM evaluation, activation patching, causal tracing, SAE workflows, TransformerLens, reproducible experiment pipelines, bootstrap resampling, CUDA / MPS.

Engineering. Python, PyTorch, Hugging Face, AWS Lambda, CloudFront, Route 53, serverless deployment.

Philosophy / Formal. Philosophy of language, metaphysics, logic, vagueness, three-valued logic, provability theory.

AI-Accelerated Research. Heavy use of frontier models for coding, experiment design, debugging, research synthesis, and analysis, with primary ownership of problem selection, methodology, evaluation, and final technical judgment.

EDUCATION

MA Philosophy, Goethe University Frankfurt

2018–2025

Grade: 1.4/1.0 (coursework completed in 12 months)

BSc Management, Philosophy and Economics, Frankfurt School of Finance & Management 2014–2018

Thesis grade: 1.1/1.0

EARLIER EXPERIENCE

Independent Consultant, Climate Risk and Sustainable Finance

2018–2024

Built quantitative climate-risk, scenario-analysis, and reporting tools for financial institutions and SMEs, including physical-risk assessment, sector-level transition-cost models, and EU Taxonomy alignment tools designed for non-technical users under regulatory constraint. Provided advisory and research support at EU and German federal sustainable-finance policy level, and developed professional training materials for sustainable-finance certification and practitioner programmes.

FS-UNEP Collaborating Centre / Frankfurt School, Analyst and Researcher

2018–2020

Authored sector analysis on green and sustainable finance in Germany, including work on TCFD adoption, EU Taxonomy alignment, and climate-related disclosure practices across banking and insurance.